Defining level geometry and layout

Every game level needs some structural formations that may consist of terrains, landscapes, platforms and buildings organized in a way to create meaningful gameplay and support game mechanics. It is important to define the level geometry and scale of the objects. This phase requires level designers to lay out the large-scale features of the map, such as hills, cities, rooms, tunnels, etc. where the players, NPC's and opponents can move around. Many game engines include tools for level editing.

Some major geometry types include:

- **BSP (Binary Space Partitioning):** This is used to generate levels that can be rendered quickly while minimizing the number of polygons that need to be redrawn every time the screen refreshes. BSP geometry is used for low detail structural geometry like cylinders, rectangles, stairs, and window frames. For more decorative objects, static mesh geometry is used.
- **Static mesh:** Used for objects that just needs to sit in the level and look pretty. They cannot be animated but can be moved, placed and reskinned in real-time. They are usually not made in the editor but are imported from popular 3D programs.
- **Terrain:** Terrain geometry is based on distance from the camera. Terrain can be set to constantly adjust its poly size based on distance from the camera. So, for example the distant hill may only be a fraction of the polygons that they will be up close. That makes Terrain much faster to use. The down side is that if you ever run at that hill, you'll notice the mountain changing shape as it gains more detail.

Determining level environment

Today video games have highly immersive environments that can leave the player gasp without a breath. Level Designers work together to decide the environmental conditions of the level: Whether it will be day or night, it will rain or flood or fire. The scoring systems, economy and time limit of each level are decided now. They decide how will starting resources be allocated in each level and how will these resources be replenished.

Placing static objects

During this step, game levels are populated with static objects, walls, terrains, buildings, trees, vegetation etc. It's vital for a level designer to under-